

Anh Ho

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EDUCATION

Stanford University

Stanford, CA

B.S. in Electrical Engineering, QuestBridge and Vida Valiente Scholar, GPA: 3.7

June 2028

Georgia Institute of Technology

Atlanta, GA

Dual Enrollment in Mathematics, GPA: 4.0

Aug. 2024- May 2025

EXPERIENCE

PCB Designer

February 2026 – Present

Stanford Sensor Mesh Network (SMesh)

Stanford, CA

- Building and installing sensor nodes to measure environmental conditions during prescribed burns.
- Designing an Altium breakout/header PCB for Heltec V3 (ESP32-S3) nodes to integrate sensors and power interfaces.
- Creating a bill of materials (BOM) and selecting connectors/components for I²C sensors and field wiring (e.g., BME680, INA series).

Embedded Software Engineer

January 2026 – Present

Stanford Solar Car Project

Stanford, CA

- Refactored C code to Rust, improving code reliability.
- Helped the team's car reduce power consumption through implementing temperature-controlled PWM fans

PROJECTS

Bare-Metal Bring-Up | *STM32, Cortex-M4, C/C++, Assembly, PS Script*

- Authored custom startup code and linker script, implementing fully bare-metal firmware with direct register manipulation, deterministic timing, and cycle-level control.
- Wrote SysTick and TIM drivers for PWM generation and timing control without HAL/RTOS, configuring a 1 kHz base to achieve 1 ms resolution.
- Debugged at register level using GDB and OpenOCD.

RTL PWM Verification Mini-Project | *Verilog, Icarus Verilog, GTKWave*

- Implemented a 4-channel PWM generator in Verilog using a 100-cycle counter and fixed duty thresholds (25/50/75/95%).
- Wrote a self-checking testbench that validates duty-cycle high counts over a full period and generates VCD waveforms for debugging.

Variable Power Supply | *Analog circuit design, oscilloscope*

- Designed and built an adjustable DC bench supply.
- Characterized regulation and ripple across loads (10k ohms–100 ohms), achieving 17.3–17.9 V output with max ripple as low as 1.9 mV.
- Validated operation up to 173 mA load current (3.0 W delivered at 17.3 V into 100 ohms) and documented results in a lab report (schematics + measurements).

Murphy | *ESP32, C++*

- Designed and assembled Wi-Fi-enabled audio playback teddy bear for long-distance communication and comfort.
- Built Firebase backend server to upload and stream audio files to device over Wi-Fi.
- Implemented embedded firmware for Wi-Fi audio streaming and GPIO-triggered playback.

TECHNICAL SKILLS

Languages: C++, Rust, Python, Verilog, Make, PowerShell Script, HTML, CSS

Tools: Altium, Linker, Startup, Icarus Verilog, GTKWave

Frameworks: arm-none-eabi-gcc, STM32CubeIDE, IAR EW, Git

CLASSES

Engineering: Circuits I (EE101A), Physics Electromagnetism (Physics 43), Physics Mechanics (Physics 41)

Computer Science: Programming Abstraction (CS106B), Harvard's Introduction to CS (CS50), Harvard's Introduction to Cybersecurity (CS50 Cybersecurity), Harvard's Introduction to AI (CS50 AI)